

Stale Resources

Resources that are no longer active, relevant, or needed but continue to occupy space in a system.

Examples:

- *Unused files or data in storage.*
- *Expired cache or session data.*
- *Outdated dependencies in software development.*
- *Abandoned cloud resources like unused VMs, databases, or storage buckets.*

Why Are Stale Resources Deleted?

1. *Optimize Performance:*
 - *Free up memory, storage, and processing power.*
 - *Speed up application and system operations.*
2. *Reduce Costs:*
 - *Eliminate unnecessary expenses in cloud or on-premises environments.*
3. *Improve Security:*
 - *Prevent vulnerabilities from outdated resources that may lack updates.*
4. *Simplify Maintenance:*
 - *Make systems easier to manage and troubleshoot by reducing clutter.*
5. *Enhance Reliability:*
 - *Prevent stale data from interfering with current operations.*

Created ec2-instance

Instances (1/3) [Info](#)

Last updated
less than a minute ago

[Connect](#)[Instance state ▼](#)[Actions ▼](#)[Launch instances](#)

Running ▼

< 1 >



☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☒	stale-instance	i-062004b38ea1bb7de	Running	t2.micro	2/2 checks passed View alarms +		us-east-1a	ec2-54-146-251-7.compute-1.amazonaws.com

i-062004b38ea1bb7de (stale-instance)

[Details](#)[Status and alarms](#)[Monitoring](#)[Security](#)[Networking](#)[Storage](#)[Tags](#)

▼ Instance summary [Info](#)

Instance ID

i-062004b38ea1bb7de

IPv6 address

—

Hostname type

IP name: ip-172-31-21-67.ec2.internal

Public IPv4 address

54.146.251.7 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-21-67.ec2.internal

Private IPv4 addresses

172.31.21.67

Public IPv4 DNS

ec2-54-146-251-7.compute-1.amazonaws.com |
[open address](#)

Created snapshot from root volume of the instance

Snapshots (1/1) [Info](#)

Owned by me ▾

Q Search

⌂

Recycle Bin

Actions ▾

Create snapshot

< 1 > ⚙

☒	Name ▾	Snapshot ID ▾	Volume size ▾	Description ▾	Storage tier ▾	Snapshot status ▾	Started
☒	cost-snapshot	snap-01d2ccc32a7a73e73	8 GiB	-	Standard	✔ Completed	2024/12/13 22:44 GM

Snapshot ID: snap-01d2ccc32a7a73e73 (cost-snapshot) ⚙ ▾

Details

Snapshot settings

Storage tier

Tags

Snapshot ID snap-01d2ccc32a7a73e73 (cost-snapshot)	Progress ✔ Available (100%)	Snapshot status ✔ Completed	Owner 600627341010
Started Fri Dec 13 2024 22:44:49 GMT+0530 (India Standard Time)	Product codes -	Fast snapshot restore -	Description -
▼ Source volume			
Volume ID	Volume size		

Created two buckets one with data another one is empty

▶ Account snapshot - updated every 24 hours

All AWS Regions

View Storage Lens dashboard

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets

Directory buckets

General purpose buckets (2)

Info

All AWS Regions

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

	Name ▲	AWS Region ▼	IAM Access Analyzer	Creation date ▼
☐	kechiya-with-data	US East (N. Virginia) us-east-1	View analyzer for us-east-1	December 13, 2024, 16:55:40 (UTC+05:30)
☐	kechiya-without-data	US East (N. Virginia) us-east-1	View analyzer for us-east-1	December 13, 2024, 22:50:34 (UTC+05:30)

kechiya-with-data Info

- Objects
- Metadata - Preview
- Properties
- Permissions
- Metrics
- Management
- Access Points

Objects (1) Info



Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 > Settings

☐	Name	Type	Last modified	Size	Storage class
☐	IMG_20211226_201600.jpg	jpg	December 13, 2024, 16:56:43 (UTC+05:30)	2.9 MB	Standard

Deleted my ec2-instance,so that snapshot will consider as a stale resource

✓ Successfully initiated termination (deletion) of i-062004b38ea1bb7de

Instances (1/3) [Info](#)

Last updated less than a minute ago

Connect

Instance state ▾

Actions ▾

Launch instances ▾

Find Instance by attribute or tag (case-sensitive)

Running ▾

< 1 >

⚙

☒	Name ✎	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☒	stale-instance	i-062004b38ea1bb7de	Shutting-d... 🔄 🔄	t2.micro	✓ 2/2 checks passed	View alarms +	us-east-1a	ec2-54-146-251-7

i-062004b38ea1bb7de (stale-instance)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary [Info](#)

Instance ID

[🔗](#) i-062004b38ea1bb7de

IPv6 address

—

Hostname type

IP name: ip-172-31-21-67.ec2.internal

Public IPv4 address

[🔗](#) 54.146.251.7 | [open address 🔗](#)

Instance state

🕒 Shutting-down

Private IP DNS name (IPv4 only)

[🔗](#) ip-172-31-21-67.ec2.internal

Private IPv4 addresses

[🔗](#) 172.31.21.67

Public IPv4 DNS

[🔗](#) ec2-54-146-251-7.compute-1.amazonaws.com | [open address 🔗](#)

Creating function in LAMBDA using python 3.11

EC2 S3 CodeDeploy CodePipeline RDS VPC IAM CloudFormation

☰ [Lambda](#) > [Functions](#) > Create function

Choose one of the following options to create your function.

☒ **Author from scratch**
Start with a simple Hello World example.

☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

cost-optimization

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.11



Architecture [Info](#)

Choose the instruction set architecture you want for your function code.

☒ x86_64

☐ arm64

Permissions [Info](#)

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

To give permission to lambda am creating ROLE in IAM

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings
- Root access management [New](#)

▼ Access reports

- Access Analyzer
 - External access
 - Unused access
 - Analyzer settings
- Credential report

cost-optimization-role-glkh7o8q

1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (1) [Info](#)

You can attach up to 10 managed policies.

Search

Filter by Type: All types

[Add permissions](#)

Attach policies

Create inline policy

☐	Policy name	Type	Attached entities
☐	AWSLambdaBasicExecutionRole-d580e0f2-...	Customer managed	1

► **Permissions boundary** (not set)

▼ **Generate policy based on CloudTrail events**

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

[Generate policy](#)

Gave limited permission to lambda to access and delete EC2 & S3

Identity and Access Management (IAM)

[IAM](#) > [Roles](#) > cost-optimization-role-e3nx5eb0

Permissions policies (3) [Info](#)

You can attach up to 10 managed policies.

[Refresh](#) [Simulate](#) [Remove](#) [Add permissions](#)

Filter by Type

☐	Policy name ?	Type	Attached entities
☐	AWSLambdaBasicExecutionRole-67b901a5...	Customer managed	1
☐	EC2	Customer inline	0
☐	S3	Customer inline	0

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

[Generate policy](#)

No requests to generate a policy in the past 7 days.

Python code to delete the stale resources in EC2 & S3

EC2 S3 CodeDeploy CodePipeline RDS VPC IAM CloudFormation

Lambda > Functions > cost-optimization

Successfully updated the function **cost-optimization**.

Untitled-1 lambda_function.py

```
lambda_function.py
9  def delete_unused_snapshots():
14      for snapshot in snapshots:
15          for snapshot_info in response['Snapshots']:
16              if not snapshot_info.get('VolumeId'): # Unused snapshot
17                  try:
18                      ec2_client.delete_snapshot(SnapshotId=snapshot_id)
19                  except ClientError as e:
20                      print(f"Error deleting snapshot {snapshot_id}: {e}")
21
22  def delete_empty_buckets():
23      response = s3_client.list_buckets()
24      buckets = response['Buckets']
25      for bucket in buckets:
26          bucket_name = bucket['Name']
27          objects = s3_client.list_objects_v2(Bucket=bucket_name)
28          if 'Contents' not in objects:
29              try:
30                  s3_client.delete_bucket(Bucket=bucket_name)
31              except ClientError as e:
32                  print(f"Error deleting bucket {bucket_name}: {e}")
33
34  # Lambda handler function
35  def lambda_handler(event, context):
36      delete_unused_snapshots()
37      delete_empty_buckets()
38      return {"statusCode": 200, "body": "Success"}
```

EXPLORER

COST-OPTIMIZATION

lambda_function.py

DEPLOY

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS

You haven't created any test events.

Create test event (Ctrl+Shift+C)

Successfully updated the function **cost-optimization**.



DEPLOY

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS

You haven't created any test events.

Create test event (Ctrl+Shift+C)



ENVIRONMENT VARIABLES

```

16 # Delete empty S3 buckets
17 delete_empty_s3_buckets(s3_client)
18
19 return {
20     'statusCode': 200,
21     'body': 'Stale snapshots and empty S3 buckets deleted.'
22 }
23
24 def delete_stale_snapshots(ec2_client, cutoff_date):
25     response = ec2_client.describe_snapshots(OwnerIds=['self'])
26
27     for snapshot in response['Snapshots']:
28         start_time = snapshot['StartTime']
29         snapshot_id = snapshot['SnapshotId']
30
31         if start_time < cutoff_date:
32             try:
33                 print(f"Deleting snapshot: {snapshot_id}, Start Time: {start_time}")
34                 ec2_client.delete_snapshot(SnapshotId=snapshot_id)
35             except Exception as e:
36                 print(f"Error deleting snapshot {snapshot_id}: {str(e)}")
37
38 def delete_empty_s3_buckets(s3_client):
39     response = s3_client.list_buckets()
    
```

Deployment successful

OUTPUT

Untitled-1 lambda_function.py X

lambda_function.py

```
9 def lambda_handler(event, context):
13     try:
42         logger.info(f"Deleted empty bucket: {bucket_name}")
43     else:
44         logger.info(f"Bucket {bucket_name} is not empty. Skipping.")
45     except Exception as e:
46         logger.error(f"Error deleting bucket {bucket_name}: {str(e)}")
47
48 except Exception as e:
49     logger.error(f"Error in Lambda function: {str(e)}")
```

PROBLEMS OUTPUT CODE REFERENCE LOG TERMINAL

Execution Results

Status: Succeeded

Test Event Name: cost-optimization

Response:

null

Function Logs:

START RequestId: e569c7f9-832e-4a95-b4d2-b675f74e02be Version: \$LATEST

[INFO] 2024-12-13T16:31:46.127Z e569c7f9-832e-4a95-b4d2-b675f74e02be

[INFO] 2024-12-13T16:31:50.168Z e569c7f9-832e-4a95-b4d2-b675f74e02be

[INFO] 2024-12-13T16:31:50.508Z e569c7f9-832e-4a95-b4d2-b675f74e02be

[INFO] 2024-12-13T16:31:50.897Z e569c7f9-832e-4a95-b4d2-b675f74e02be

[INFO] 2024-12-13T16:31:51.179Z e569c7f9-832e-4a95-b4d2-b675f74e02be

END RequestId: e569c7f9-832e-4a95-b4d2-b675f74e02be

REPORT RequestId: e569c7f9-832e-4a95-b4d2-b675f74e02be Duration: 5286.04 ms Billed Duration: 5287 ms Memory Size: 128 MB Max Memory Used: 100 MB Init Duration: 310.19 ms

Request ID: e569c7f9-832e-4a95-b4d2-b675f74e02be

Found credentials in environment variables.

Starting deletion of unused EC2 snapshots...

Starting deletion of empty S3 buckets...

Bucket kechiya-with-data is not empty. Skipping.

Deleted empty bucket: kechiya-without-data

Snapshot deleted

Snapshots [Info](#)

Owned by me ▼

🔍 Search

🔄

Recycle Bin

Actions ▼

Create snapshot

< 1 >

⚙️

☐	Name ▼	Snapshot ID ▼	Volume size ▼	Description ▼	Storage tier ▼	Snapshot status ▼	Started
You currently have no snapshots in this Region.							

Only data bucket is available,empty bucket deleted

EC2S3CodeDeployCodePipelineRDSVPCIAMCloudFormation

Amazon S3> Buckets

Account snapshot - updated every 24 hoursAll AWS RegionsView Storage Lens dashboard

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. Learn more

General purpose bucketsDirectory buckets

General purpose buckets (1)InfoAll AWS Regions

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >⚙

Name	AWS Region	IAM Access Analyzer	Creation date
☐ kechiya-with-data	US East (N. Virginia) us-east-1	View analyzer for us-east-1	December 13, 2024, 16:55:40 (UTC+05:30)

Risks of Not Deleting Stale Resources

1. System Inefficiency:

- Slower system performance due to excessive resource utilization.

2. Higher Costs:

- Continued payment for unused cloud or IT resources.

3. Security Vulnerabilities:

- Stale resources may include outdated or unpatched components.

4. Compliance Issues:

- Regulatory violations for storing outdated or irrelevant data (e.g., GDPR).